

STC 122 - Practical 3 instructions

Instructions

File and Submission Requirements

- Answer the questions that follow and save all your code in a single R script.
- Name the script **Practical3.R**, where **Practical3** is the filename and **.R** is the file extension.
- Once you have completed your online submission, ensure the script is uploaded to clickUP using the correct file name, **Practical3.R**.
- Ensure all variables are named correctly. **Remember that variable names are case sensitive.**
- Ensure your code does not contain any syntax errors.
- Any code that is commented out will be treated as rough work and will not be marked.

Preparing for the Online Assessment

- The questions in this document are intended to prepare you for the online submission. Make sure you are comfortable with all questions here, as well as the content covered in the lecture notes, before attempting the online assessment.
- You will have **only ONE attempt** at the online submission, and it must be completed in **ONE sitting**. **You cannot save your progress and continue later. If you exit the attempt before submitting all your answers, you will be unable to resume and will forfeit the marks for the practical.**
- It is your responsibility to ensure you have a stable and sufficient internet connection before starting the online submission.

Academic Requirements

- You must include **detailed** explanations of your working as comments in your code, for **every** question in the assignment.
- You must also cite any resources you made use of while completing the assignment. Refer to the document `commenting_and_citing_in_your_code.pdf` on clickUP for guidance.
- You must complete the Declaration of Generative AI Use form after finishing your online submission. **Failure to submit this form will result in a mark of zero for the practical.**

Deadline

- The deadline for this practical is **Friday, 2 October 2026 at 23:59**.

Questions

- This assignment is based on the online Racer game from the Statistics Department at Grinnell College, Iowa, USA. It is not necessary to play the game in order to do this assignment, but see **Stat2games** to access the game and for more details regarding the game.
- Where applicable, answer the questions below by typing the appropriate **code** in the R script template provided on ClickUP. Some questions are theoretical and no coding is needed to answer those questions.
- For this assignment you should use the dataset in the file `RacingGameData.csv` that is available on clickUP. The dataset contains the results of multiple races completed in the above-mentioned online racing game. The variables in the dataset are defined as follows:
 - Track: the track on which the race occurred.
 - Engine: the engine of the car used to complete the race.
 - FinishTime: the time, in seconds, to complete the race.

Question 1

Consider the continuous variable `FinishTime` for all races in the dataset. Create an indicator variable called `speed` indicating whether the car was completed within 25 seconds. Assign it the value "fast" if the race was completed in less than 25 seconds, and "slow" if it took 25 seconds or more.

HINT: Use the `ifelse` function in R.

Question 2

Test the hypothesis that the proportion of races where the race was completed in less than 25 seconds is less than 55%.

- a) Select the option with the correct null hypothesis. Assign the correct option to a variable called `Q2a`.

For example, if the correct option is option E, assign "E" to a variable called `Q2a`:

```
Q2a <- "E"
```

A.	$H_0 : p > 0.55$
B.	$H_0 : \hat{p} = 0.55$
C.	$H_0 : p = 0.55$
D.	$H_0 : \hat{p} < 0.55$

- b) Select the option with the correct alternative hypothesis. Assign the correct option to a variable called Q2b.

For example, if the correct option is option E, assign "E" to a variable called Q2b:

Q2b <- "E"

A.	$H_A : p > 0.55$
B.	$H_A : \hat{p} = 0.55$
C.	$H_A : p = 0.55$
D.	$H_A : \hat{p} < 0.55$

- c) Calculate the point estimate for the proportion of fast races. Assign your answer to a variable called Q2c.
- d) Calculate the standard error required to calculate the test statistic value for the hypothesis test. Assign your answer to a variable called Q2d.
- e) Calculate the test statistic value. Assign your answer to a variable called Q2e.
- f) Given that the p -value is 0.04, give the appropriate decision and conclusion for the hypothesis test. Assign the correct option to a variable called Q2f.

For example, if the correct option is option E, assign "E" to a variable called Q2f:

Q2f <- "E"

A.	Do not reject the null hypothesis at a 5% level of significance and conclude that the proportion of races completed in less than 25 seconds is 55%.
B.	Reject the null hypothesis at a 5% level of significance and conclude that the proportion of races where the race was completed in less than 25 seconds is 55%.
C.	Do not reject the null hypothesis at a 5% level of significance and conclude that the proportion of races where the race was completed in less than 25 seconds is not 55%.
D.	Reject the null hypothesis at a 5% level of significance and conclude that the proportion of races where the race was completed in less than 25 seconds is not 55%.

Question 3

Let p_1 be the proportion of races where the race was completed in less than 25 seconds with a car that has a Nightingale engine, and let p_2 be the proportion of races where the race was completed in less than 25 seconds with a car that has a Gauss engine. A 90% confidence interval must be calculated for $p_1 - p_2$.

- a) Calculate $\hat{p}_1 - \hat{p}_2$. Assign your answer to a variable called Q3a.

- b) Use bootstrapping to obtain simulated differences between the two sample proportions. Use a seed value of 10 and 1000 bootstrap samples. Assign your answer to a variable called Q3b.
- c) Calculate the lower bound of the 90% confidence interval using the bootstrap percentile method. Assign your answer to a variable called Q3c.
- d) Calculate the upper bound of the 90% confidence interval using the bootstrap percentile method. Assign your answer to a variable called Q3d.

HINT: Use the `quantile` function.

Question 4

Let p_1 be the proportion of races where the race was completed in less than 25 seconds on the straight track, and let p_2 be the proportion of races where the race was completed in less than 25 seconds on the oval track.

Test the following hypothesis:

$$H_0 : p_1 - p_2 = 0.25$$

$$H_A : p_1 - p_2 \neq 0.25$$

- a) Calculate the standard error required to calculate the test statistic value for the hypothesis test. Assign your answer to a variable called Q4a.
- b) Calculate the test statistic value. Assign your answer to a variable called Q4b.
- c) Calculate the p -value. Assign your answer to a variable called Q4c.
- d) Based on the p -value calculated in Question 4(c), provide a decision for the test. Assign the correct option to a variable called Q4d.

For example, if the correct option is option E, assign "E" to a variable called Q4d:

```
Q4d <- "E"
```

A. Do not reject the null hypothesis at a 5% level of significance.
B. Reject the null hypothesis at a 5% level of significance.

Question 5

Use a chi-square test for independence to determine whether the event that the race is completed in less than 25 seconds is independent of the type of engine used.

- a) Select the option with the correct null hypothesis. Assign the correct option to a variable called Q5a.

For example, if the correct option is option E, assign "E" to a variable called Q5a:

```
Q5a <- "E"
```

A. H_0 : The event that the race is completed in less than 25 seconds is independent of the engine used.
B. H_0 : The event that the race is completed in less than 25 seconds is dependent on the engine used.

- b) Calculate the test statistic. Assign your answer to a variable called Q5b.
- c) Calculate the p -value. Assign your answer to a variable called Q5c.
- d) Give the expected count of races completed with a car that has a Nightingale engine finishing the race in less than 25 seconds. Assign your answer to a variable called Q5d.